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The Editor-in-Chief
Reliability Engineering & System Safety
Elsevier

Dear Editor,

We submit for your consideration the manuscript entitled “BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm” as a Research Paper.

The manuscript proposes a quantitative-coupled BowTie procedure for design-stage reliability assessment of the valve train of a high-performance spark-ignition cylinder head, organised around a top event of catastrophic significance for high-rpm engines: the loss of follower contact between valve and cam at 10,000 rpm. Beyond the qualitative bow diagram, each preventive barrier is associated with a deterministic indicator computed from the analytical sizing of the intake valve and the dual chrome–silicon valve springs (dynamic follower-contact ratio, natural-frequency margin against cam harmonics, coil-bind clearance, and Goodman fatigue factor of safety), with literature-anchored pass criteria. A Monte Carlo refinement under ASTM A877/A877M tolerances ($N = 2 \times 10^5$ trials) converts the deterministic at-risk classification into probabilities of barrier compliance failure under manufacturing-realistic dispersion, and surfaces a third marginal barrier that the nominal-point analysis would have missed. The procedure is qualitatively validated against four documented valve-train and engine-component failure cases from the recent literature, and a concrete experimental validation roadmap — spring-rig modal verification, motored-engine speed sweep, and inner-spring fatigue campaign, with explicit acceptance criteria — is specified for empirical confirmation.

We believe *Reliability Engineering & System Safety* is the right venue for this work for three reasons. First, the BowTie methodology is a central thread of the journal: from the seminal contributions of Khakzad, Khan and Amyotte (RESS 2012, DOI 10.1016/j.res.2012.04.003) to recent extensions by Vileiniskis and Remenyte-Priscott (RESS 2017), Taleb-Berrouane and colleagues (RESS 2021), Xie and colleagues (RESS 2021), and Sezer and colleagues (RESS 2023), RESS has consistently published the methodological advances on which the present manuscript builds. Second, the quantitative coupling between barrier and deterministic indicator addresses a gap explicitly identified in the BowTie literature — how to convert a qualitative diagram into a verifiable engineering check at the design stage — and is therefore aligned with the journal’s mission of advancing methods for the safety and reliability of complex engineered systems. Third, the application domain (valve-train dynamics at extreme rotational speed in a passenger-car platform with a wide installed base in Brazil) extends BowTie risk analysis from its classical chemical/process-safety home into the mechanical-reliability domain, where the literature is currently thinner and where the proposed deterministic-indicator coupling has immediate engineering value.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All four authors have approved the submission, and the authors declare no competing financial interests or personal relationships that could have influenced the work. No funding was received for this research. No new experimental data were generated; the Monte Carlo script is provided as supplementary material.

We respectfully suggest the following independent reviewers as potentially appropriate. None has co-authored work with the present authors, and none is affiliated with the Universidade Católica de Petrópolis (UCP) or the Pontifícia Universidade Católica do Rio de Janeiro (PUC-Rio). Institutional contact details can be located on each researcher’s institutional page.

1. **Prof. Valerio Cozzani** — Department of Civil, Chemical, Environmental and Materials Engineering (DICAM), University of Bologna, Italy. Leading authority on safety-barrier performance, LOPA, and Natech-cascade analysis; multiple recent RESS publications on barrier degradation (e.g. RESS 2021, DOI 10.1016/j.res.2021.107634). Relevance: barrier-based reasoning is the central methodological device of the present manuscript.
2. **Prof. Chia-Fen Chi** — Department of Industrial Management, National Taiwan University of Science and Technology (NTUST), Taipei, Taiwan. Recent RESS work on FMEA classification of passenger-vehicle recalls (RESS 2020, DOI 10.1016/j.res.2020.106929). Relevance: the present manuscript explicitly compares the quantitative-coupled BowTie against FMEA/RPN, including an AIAG-VDA 2019 scoring of the six identified threats, which is directly within Prof. Chi’s published expertise.
3. **Prof. Saeed Khodaygan** — Department of Mechanical Engineering, Sharif University of Technology, Tehran, Iran. Recent RESS work on Bayesian-reliability-based multi-objective tolerance design of mechanical assemblies (RESS 2021, DOI 10.1016/j.res.2021.107748). Relevance: the manuscript’s Monte Carlo refinement under ASTM A877/A877M wire-and-strength tolerances sits squarely in the mechanical-reliability-under-tolerance domain.
4. **Prof. Frank Guldenmund** — Section Safety and Security Science, Faculty of Technology, Policy and Management, Delft University of Technology, Netherlands. Co-author of the canonical BowTie review (de Ruijter and Guldenmund, *Safety Science*, 2016, DOI 10.1016/j.ssci.2016.03.001) cited in our manuscript; senior authority on the qualitative formulation of the BowTie diagram.
5. **Prof. Gabriele Landucci** — Department of Civil and Industrial Engineering, University of Pisa, Italy. Co-author of multiple RESS papers on safety-barrier performance under domino-effect and Natech scenarios (e.g. RESS 2020, DOI 10.1016/j.res.2019.106597). Relevance: barrier-performance assessment under realistic dispersion is the methodological core of our Monte Carlo layer.

The final selection of reviewers is, of course, entirely at the editor’s discretion.

We look forward to your editorial decision and remain available for any clarification.

Yours sincerely,

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